Waiting Time for a Dice Pattern

Absorbing Markov Chains and First-Step Analysis

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 1

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1 The Problem

> Roll a fair six-sided die repeatedly until the consecutive pair $(2,3)$ appears for the first time. What is the expected number of rolls?

We model the process as an **absorbing Markov chain** whose states record how much of the pattern has already been matched.

2 Constructing the Chain

2.1 State Space

The die rolls themselves are i.i.d. and carry no memory, but *progress toward the pattern* does. Three states suffice:

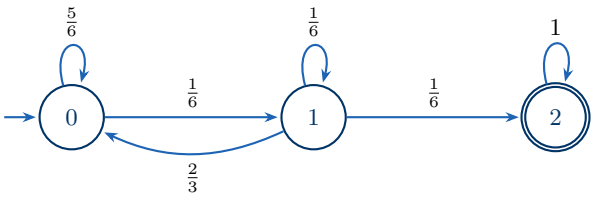
State	Name	Meaning
0	Start	No progress: we have just begun, or the last roll was not a 2
1	Partial	The last roll was a 2; a 3 next completes the pattern
2	Success	The pattern $(2,3)$ has occurred — absorbing, we stop

2.2 Transitions

From state 0, a 2 (probability $1/6$) advances to state 1 and anything else keeps us in state 0. From state 1, a 3 absorbs into state 2; a 2 leaves us in state 1, since the new 2 is itself a fresh partial match; and any of $1,4,5,6$ destroys the progress and returns us to state 0. Hence

$$P = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} & 0 \\ \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 1 \end{pmatrix}, \qquad P_{ij} = P(\text{state } j \text{ next} \mid \text{state } i \text{ now}).$$

The self-loop at state 1 is the whole subtlety. Rolling a 2 while already holding a 2 is neither progress nor failure — it resets the partial match to itself. Getting this edge wrong is the usual error in pattern-waiting problems.



3 First-Step Analysis

Let E_i be the expected number of further rolls to absorption starting from state i. Conditioning on the next roll and using the tower property gives one equation per transient state — the 1 accounts for the roll just taken:

$$E_0 = 1 + \tfrac{5}{6}E_0 + \tfrac{1}{6}E_1, \qquad E_1 = 1 + \tfrac{2}{3}E_0 + \tfrac{1}{6}E_1 + \tfrac{1}{6}\cdot 0, \qquad E_2 = 0.$$

The first equation collapses to $\frac{1}{6}E_0 = 1 + \frac{1}{6}E_1$, that is

$$E_0 = 6 + E_1. \tag{1}$$

The second gives $\frac{5}{6}E_1 = 1 + \frac{2}{3}E_0$, that is $E_1 = \frac{6}{5} + \frac{4}{5}E_0$. Substituting,

$$E_1 = \tfrac{6}{5} + \tfrac{4}{5}(6 + E_1) = 6 + \tfrac{4}{5}E_1 \quad\Longrightarrow\quad \tfrac{1}{5}E_1 = 6 \quad\Longrightarrow\quad E_1 = 30,$$

and therefore $E_0 = 36$.

Answer. The expected number of rolls to first see $(2,3)$ is $\boxed{36}$; from a state where a 2 has just been rolled, 30 further rolls are expected.

4 The Fundamental Matrix

4.1 Canonical Form

Writing the chain with transient states first,

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}, \qquad Q = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{6} \end{pmatrix}, \qquad R = \begin{pmatrix} 0 \\ \frac{1}{6} \end{pmatrix},$$

the **fundamental matrix** is $N = (I - Q)^{-1}$ and the vector of expected absorption times is $\mathbf{t} = N\mathbf{1}$. The entry N_{ij} has its own meaning: the expected number of visits to transient state j before absorption, starting from i. Summing the row therefore counts total steps.

4.2 Computation

$$I - Q = \begin{pmatrix} \frac{1}{6} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{5}{6} \end{pmatrix}, \qquad \det(I - Q) = \tfrac{5}{36} - \tfrac{4}{36} = \tfrac{1}{36},$$

so, using $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1} = \frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$,

$$N = 36\begin{pmatrix} \frac{5}{6} & \frac{1}{6} \\ \frac{2}{3} & \frac{1}{6} \end{pmatrix} = \begin{pmatrix} 30 & 6 \\ 24 & 6 \end{pmatrix}, \qquad \mathbf{t} = N\mathbf{1} = \begin{pmatrix} 36 \\ 30 \end{pmatrix},$$

confirming $E_0 = 36$ and $E_1 = 30$. The matrix route scales: for a pattern of length m the chain has m transient states and the same two lines of linear algebra apply, whereas the substitution method becomes unwieldy.

5 Why the Answer Depends on Self-Overlap

The value 36 is specific to the structure of $(2,3)$, not to its probability. It arises because $(2,3)$ cannot overlap itself: no proper suffix of the pattern is also a prefix, so a failed match never leaves us partially matched at a shorter length.

5.1 The Overlapping Case $(1,1)$

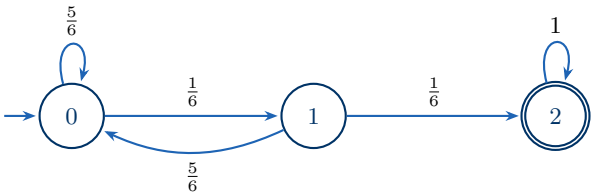
Now wait instead for two consecutive 1s. The state space is unchanged in form — state 0 is no progress, state 1 is "the last roll was a 1", state 2 is absorption — but one row of the transition matrix changes, and that is enough to change the answer.

Where the two chains diverge. In the pattern $(2,3)$ the symbol that *starts* the pattern (a 2) differs from the symbol that *completes* it (a 3). Sitting in state 1 and rolling a 2 therefore does neither: it discards the old partial match and installs a new one of the same length, so the chain stays at state 1. In $(1,1)$ the starting and completing symbols are the same. Rolling a 1 from state 1 completes the pattern and absorbs; there is no roll that holds us at state 1. Every other face sends us back to state 0.

From state 1, roll…	Pattern $(2,3)$		Pattern $(1,1)$	
	Faces	Go to	Faces	Go to
completes the pattern	3	state 2 (abs.)	1	state 2 (abs.)
restarts a partial match	2	state 1	—	—
destroys all progress	1, 4, 5, 6	state 0	2, 3, 4, 5, 6	state 0

So for $(1,1)$ the transition matrix and diagram are

$$P = \begin{pmatrix} \frac{5}{6} & \frac{1}{6} & 0 \\ \frac{5}{6} & 0 & \frac{1}{6} \\ 0 & 0 & 1 \end{pmatrix},$$



— identical to the $(2,3)$ diagram except that the self-loop at state 1 is gone and its probability has been folded into the fallback edge, which now carries $5/6$ instead of $2/3$.

First-step analysis. The equations are

$$E_0 = 1 + \tfrac{5}{6}E_0 + \tfrac{1}{6}E_1, \qquad E_1 = 1 + \tfrac{5}{6}E_0 + \tfrac{1}{6}\cdot 0.$$

The first is unchanged from before and still gives $E_0 = 6 + E_1$. Substituting it into the second,

$$E_1 = 1 + \tfrac{5}{6}(6 + E_1) = 6 + \tfrac{5}{6}E_1 \quad\Longrightarrow\quad \tfrac{1}{6}E_1 = 6 \quad\Longrightarrow\quad E_1 = 36, \qquad E_0 = 42.$$

The fundamental matrix agrees: with $Q = \left(\begin{smallmatrix} 5/6 & 1/6 \\ 5/6 & 0 \end{smallmatrix}\right)$ we get $\det(I - Q) = \frac{1}{6} - \frac{5}{36} = \frac{1}{36}$ and $N = \left(\begin{smallmatrix} 36 & 6 \\ 30 & 6 \end{smallmatrix}\right)$, so $\mathbf{t} = N\mathbf{1} = (42,\, 36)^\top$.

> Compare the two: $E_1 = 30$ for $(2,3)$ but $E_1 = 36$ for $(1,1)$. Holding a partial match is worth more in the non-overlapping case, because a 2 that fails to complete $(2,3)$ still leaves the chain at state 1, whereas any roll that fails to complete $(1,1)$ throws away the progress entirely. The self-loop is a cushion, and $(1,1)$ has none.

5.2 General Rule

The two chains differ only in whether state 1 has a self-loop, and that single edge is what separates 36 from 42. The pattern generalises: any two-symbol pattern (a,b) with $a \neq b$ takes 36 rolls in expectation, and any repeated pattern (a,a) takes 42, even though both have probability $1/36$. Self-overlap, not probability, is what makes a pattern slow to arrive.

6 Where This Shows Up

The same three-step recipe — define states as partial progress, write first-step equations, invert $I - Q$ — answers any "expected time until a pattern appears" question: runs of consecutive defects in quality control, motif positions in a genome, pattern matching in a data stream, and, in our setting, the expected time until a specified run of signs or regime labels appears in a return series. The last of these is the practical reason to be careful: a

trading rule triggered by a pattern will fire far less often than $1/P(\text{pattern})$ suggests when the pattern overlaps itself, and backtests that ignore this systematically misjudge how many independent signals the data actually contains.